1. Solve x^2 - 4 = 0.

Response: x = 2

Synthetic teacher mark: 1/2
Correct answer: x = 2 or x = -2
Feedback: Only the positive root is written.

2. Solve (x - 2)(x + 3) = 0.

Response: x = 2

Synthetic teacher mark: 1/2
Correct answer: x = 2 or x = -3
Feedback: The second factor's root is missing.

3. Expand (x + 2)(x - 1).

Response: x^2 + 3x - 2

Synthetic teacher mark: 1/2
Correct answer: x^2 + 1x - 2
Feedback: The written middle coefficient is incorrect.

SYNTHETIC - assistant draft labels; maintainer review required

4. Find the vertex of y = (x - 2)^2 + 3.

Response: (2, 0)

Synthetic teacher mark: 1/2
Correct answer: (2, 3)
Feedback: The vertex's vertical coordinate is incorrect.

SYNTHETIC - assistant draft labels; maintainer review required

5. Find the axis of symmetry of y = (x + 2)^2 - 1.

Response: x = 2

Synthetic teacher mark: 1/2

Correct answer: x = -2

Feedback: The written axis has the wrong sign.

6. Find the discriminant of x^2 + 4x + 2 = 0.

Response: D = (4)^2 + 4(2) = 24

Synthetic teacher mark: 1/2
Correct answer: D = 8
Feedback: The response adds 4ac instead of subtracting it.

7. Find the y-intercept of y = x^2 - 3x + 2.

Response: (2, 0)

Synthetic teacher mark: 1/2
Correct answer: (0, 2)
Feedback: The coordinates written for the y-intercept are reversed.

SYNTHETIC - assistant draft labels; maintainer review required

8. Factor 2x^2 - 4x completely.

Response: 2x(x + 2)

Synthetic teacher mark: 1/2
Correct answer: 2x(x - 2)
Feedback: The second term inside the factor has the wrong sign.

9. Classify the turning point of y = -(x - 2)^2 + 4.

Response: Minimum at (2, 4)

Synthetic teacher mark: 1/2
Correct answer: Maximum at (2, 4)
Feedback: The response labels the turning point as a minimum.

10. Complete the square: x^2 + 4x + 7.

Response: (x + 2)^2

Synthetic teacher mark: 1/2
Correct answer: (x + 2)^2 + 3
Feedback: The constant adjustment is missing from the written expression.

11. Solve x^2 - 9 = 0.

Response: x = 3

Synthetic teacher mark: 1/2
Correct answer: x = 3 or x = -3
Feedback: Only the positive root is written.

12. Solve (x - 3)(x + 4) = 0.

Response: x = 3

Synthetic teacher mark: 1/2
Correct answer: x = 3 or x = -4
Feedback: The second factor's root is missing.

13. Expand (x + 3)(x - 1).

Response: x^2 + 4x - 3

Synthetic teacher mark: 1/2
Correct answer: x^2 + 2x - 3
Feedback: The written middle coefficient is incorrect.

SYNTHETIC - assistant draft labels; maintainer review required

14. Find the vertex of y = (x - 3)^2 + 4.

Response: (3, 0)

Synthetic teacher mark: 1/2
Correct answer: (3, 4)
Feedback: The vertex's vertical coordinate is incorrect.

SYNTHETIC - assistant draft labels; maintainer review required

15. Find the axis of symmetry of y = (x + 3)^2 - 1.

Response: x = 3

Synthetic teacher mark: 1/2
Correct answer: x = -3
Feedback: The written axis has the wrong sign.

16. Find the discriminant of x^2 + 5x + 3 = 0.

Response: D = (5)^2 + 4(3) = 37

Synthetic teacher mark: 1/2
Correct answer: D = 13
Feedback: The response adds 4ac instead of subtracting it.

SYNTHETIC - assistant draft labels; maintainer review required

17. Find the y-intercept of y = x^2 - 4x + 3.

Response: (3, 0)

Synthetic teacher mark: 1/2
Correct answer: (0, 3)
Feedback: The coordinates written for the y-intercept are reversed.

18. Factor 3x^2 - 9x completely.

Response: 3x(x + 3)

Synthetic teacher mark: 1/2
Correct answer: 3x(x - 3)
Feedback: The second term inside the factor has the wrong sign.

19. Classify the turning point of y = -(x - 3)^2 + 9.

Response: Minimum at (3, 9)

Synthetic teacher mark: 1/2
Correct answer: Maximum at (3, 9)
Feedback: The response labels the turning point as a minimum.

20. Complete the square: x^2 + 6x + 12.

Response: (x + 3)^2

Synthetic teacher mark: 1/2
Correct answer: (x + 3)^2 + 3
Feedback: The constant adjustment is missing from the written expression.

21. Solve x^2 - 16 = 0.

Response: x = 4

Synthetic teacher mark: 1/2
Correct answer: x = 4 or x = -4
Feedback: Only the positive root is written.

22. Solve (x - 4)(x + 5) = 0.

Response: x = 4

Synthetic teacher mark: 1/2
Correct answer: x = 4 or x = -5
Feedback: The second factor's root is missing.

23. Expand (x + 4)(x - 1).

Response: x^2 + 5x - 4

Synthetic teacher mark: 1/2
Correct answer: x^2 + 3x - 4
Feedback: The written middle coefficient is incorrect.

SYNTHETIC - assistant draft labels; maintainer review required

24. Find the vertex of y = (x - 4)^2 + 5.

Response: (4, 0)

Synthetic teacher mark: 1/2
Correct answer: (4, 5)
Feedback: The vertex's vertical coordinate is incorrect.

25. Find the axis of symmetry of y = (x + 4)^2 - 1.

Response: x = 4

Synthetic teacher mark: 1/2
Correct answer: x = -4
Feedback: The written axis has the wrong sign.

26. Find the discriminant of x^2 + 6x + 4 = 0.

Response: D = (6)^2 + 4(4) = 52

Synthetic teacher mark: 1/2
Correct answer: D = 20
Feedback: The response adds 4ac instead of subtracting it.

SYNTHETIC - assistant draft labels; maintainer review required

27. Find the y-intercept of y = x^2 - 5x + 4.

Response: (4, 0)

Synthetic teacher mark: 1/2
Correct answer: (0, 4)
Feedback: The coordinates written for the y-intercept are reversed.

28. Factor 4x^2 - 16x completely.

Response: 4x(x + 4)

Synthetic teacher mark: 1/2
Correct answer: 4x(x - 4)
Feedback: The second term inside the factor has the wrong sign.

29. Classify the turning point of y = -(x - 4)^2 + 16.

Response: Minimum at (4, 16)

Synthetic teacher mark: 1/2
Correct answer: Maximum at (4, 16)
Feedback: The response labels the turning point as a minimum.

30. Complete the square: x^2 + 8x + 19.

Response: (x + 4)^2

Synthetic teacher mark: 1/2
Correct answer: (x + 4)^2 + 3
Feedback: The constant adjustment is missing from the written expression.